Broadcast Technology UG1

Lighting and its Control

Luminaires

Lantern, Instrument

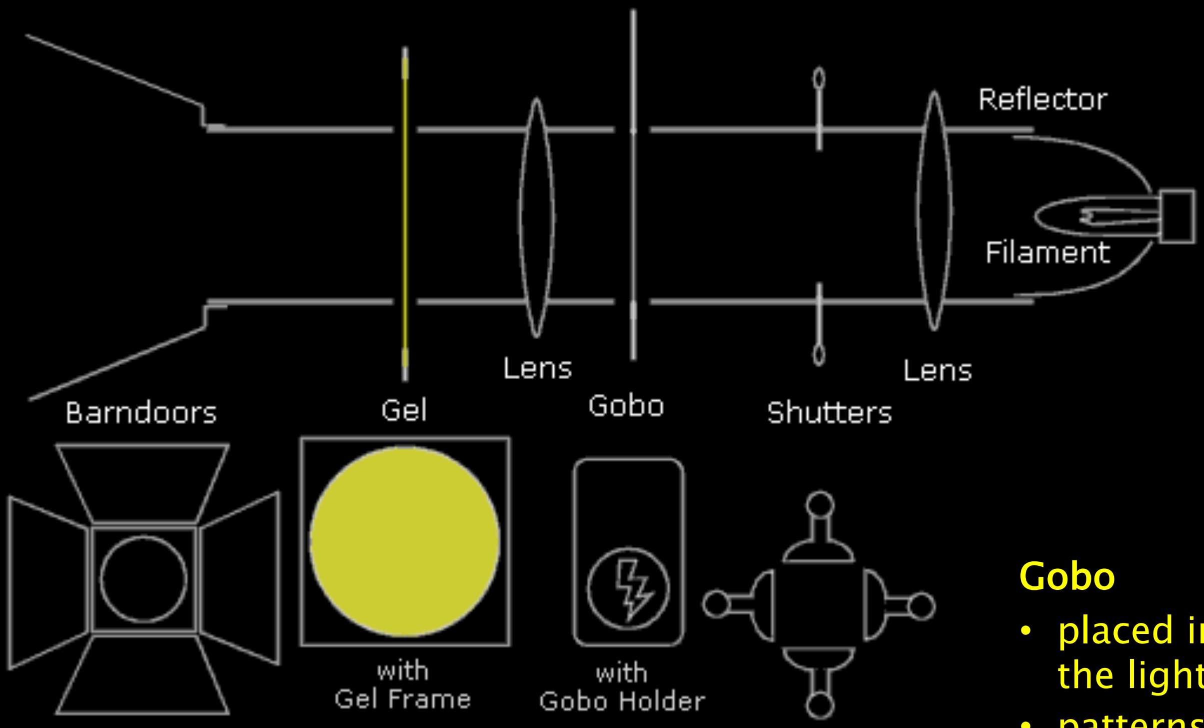
Housing
- metal or plastic
- prevents light spilling in unwanted directions
- to reduce heat, increase efficiency

- originally sheet steel or aluminium
- increasingly die cast metal
    - allows single, light-weight body
    - economical to produce and use
- some made from plastic

Lens Opening
- lens often used to control the beam
- reflector

Luminaires

(not all of these features may be present in any given lamp)



Reflector

- directs light in a specific direction
- typically circular or oval to form a beam of light

Lens

- typically – curved (plano-convex) lens
- tend to be thick and heavy
- option – Fresnel lens

Gobo

- placed inside some lights to shape the light into patterns
- patterns, logos, text, shapes

Shutters

- series of metal plates to shape the beam
- fully retracted – beam is circular
- pushed in – beam can be square or rectangular

Barndoors

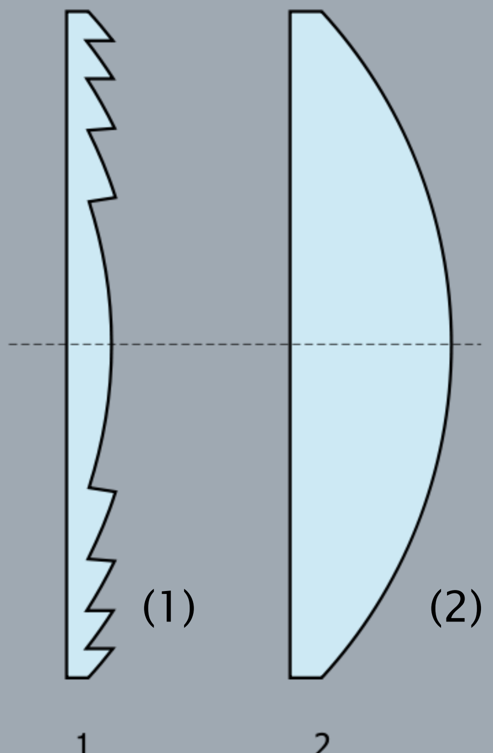
- flaps – like shutters but not as precise, generally used to stop spilled light

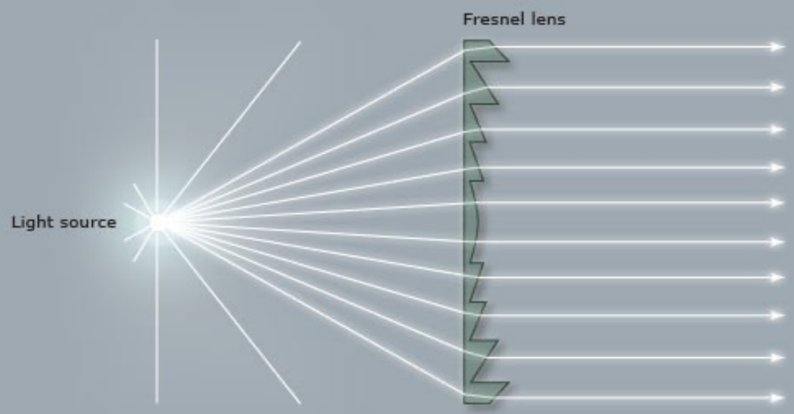
Gel (or colour filter)

- colour correction
- create mood
- coloured sheet of plastic fixed to the front inside a gel frame

http://people.bath.ac.uk/su2bc/infoguides/smc/basiclighting/index.shtml

Fresnel Lens





Fresnel Lens (1)
- focuses light by using a series of separate sections

Conventional lens (2)
- single, large lens, similar magnification power
- significantly thicker, more expensive, harder to build

Incandescent (Tungsten) Lamp

Incandescence

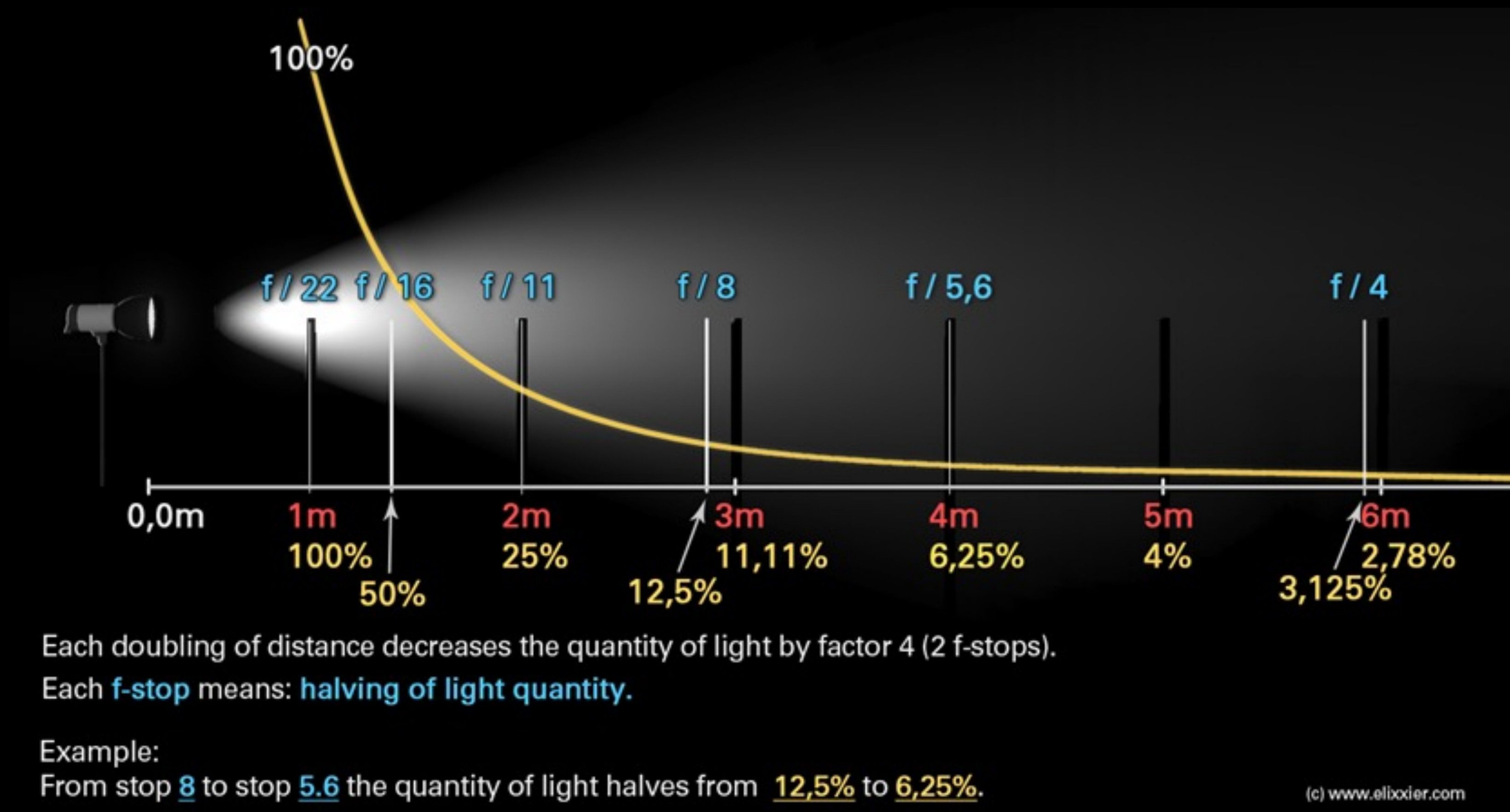
- heated tungsten filament
- black body radiation
- enclosed in a glass bulb
  - vacuum or an inert gas to prevent oxidation of the filament

- wide range of sizes and voltages
- require no external regulating equipment
- low manufacturing cost
- AC or DC
- widely used – household and commercial
  - being phased out, usually for energy saving alternatives

Very Inefficient:

- radiate mostly infrared radiation
- Incandescent lamp ~90% power consumed is emitted as heat

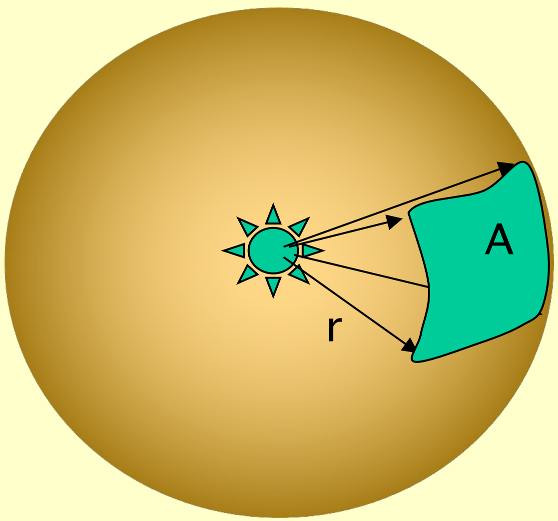
Propagation of Light (and Sound): Inverse Square Law

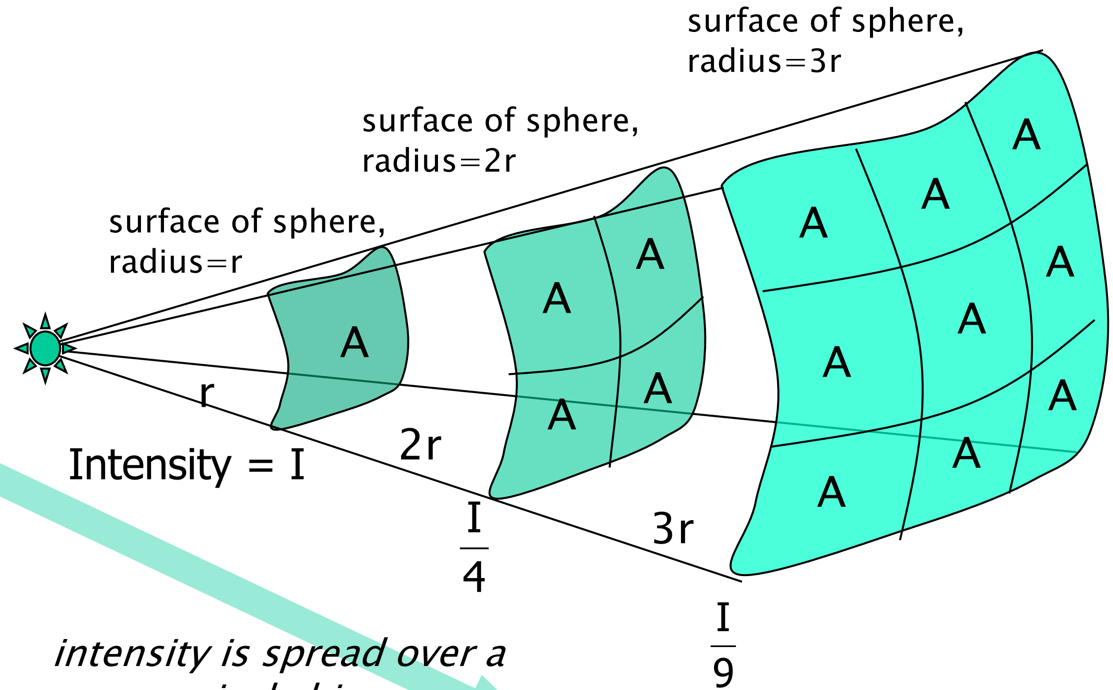


Between the 1st and second meter the brightness (intensity) has already decreased by 75% to only 25%
= 2 f-stops

Inverse Square Law

e.g. the reduction of light, sound intensity further from the source





intensity is spread over a progressively bigger area

Sphere radius $= r$

Sphere surface area $= 4\pi r^2$

Source Power $= P$

$\therefore$ intensity at surface of sphere

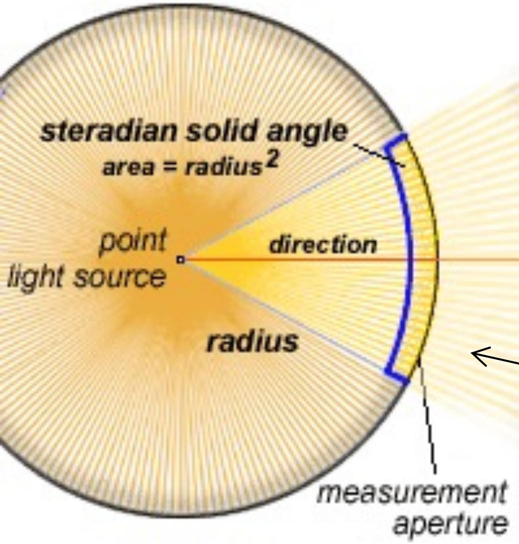
$$= I = \frac{\text{Power}}{\text{Area}} = \frac{P}{4\pi r^2}$$

<u>Intensity (Energy):</u>

$$\left.\begin{aligned} &\text{at } r = I \\ &\text{at } 2r = \frac{I}{4} = \frac{I}{2^2} \\ &\text{at } 3r = \frac{I}{9} = \frac{I}{3^2} \end{aligned}\right\} \text{at } nr = \frac{I}{n^2}$$

Lumens, Luminous Efficacy

Luminous Flux
- visible light energy emitted, the power of light perceived by the human eye
- i.e. the varying sensitivity of the human eye to different wavelengths of light



Radiant flux
- total power of light emitted, Watts

Luminous Intensity
- quantity of luminous flux emitted in a given direction per *solid angle*

Luminous efficacy – *a ratio …*

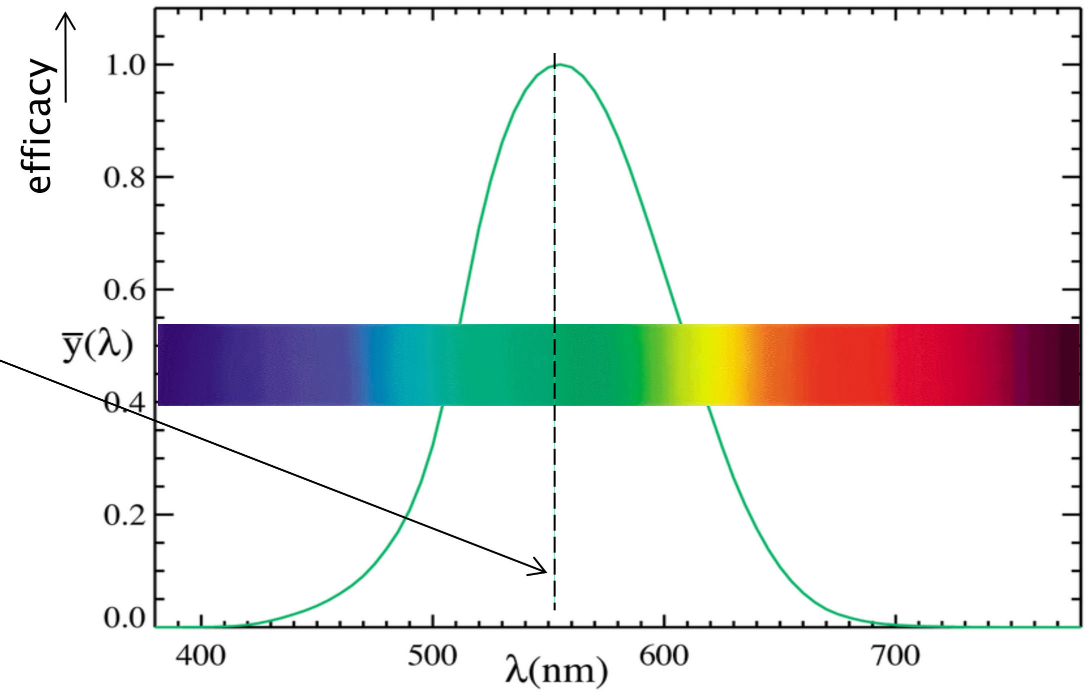
Efficacy = "Effectiveness"

Visible light energy emitted (Luminous Flux) in ***Lumens***	:	Total power input to the lamp in ***Watts***

… a measure of how well the distribution of energy matches the perception of the eye

Lumens, Luminous Efficacy

Maximum **efficacy possible**
is **683 lm/W**
for monochromatic green light
at 555 nanometres wavelength
... the peak sensitivity of the
human eye



For "white light"
maximum luminous efficacy is
~240 lm/W
... the human eye perceives
many different mixtures of
visible light as "white"

Tungsten / Halogen Lamps (aka Quartz Halogen)

- tungsten filament “hot lights”
- general use in film market up to 20kW
- mobile – typ up to 2kW
- for DV cameras ~1kW usually ok

Colour – Nominally 3000 – 3200K

Advantages
- Cheap, uncomplicated
- no need for an additional starter circuits

Disadvantages
- Hot
- Bulb life often short; often expensive bulbs
- for daylight, a colour correction filter required
    - loses half the light output
- Inefficient, Lumen-Per-Watt
    - least efficient technology on the market today

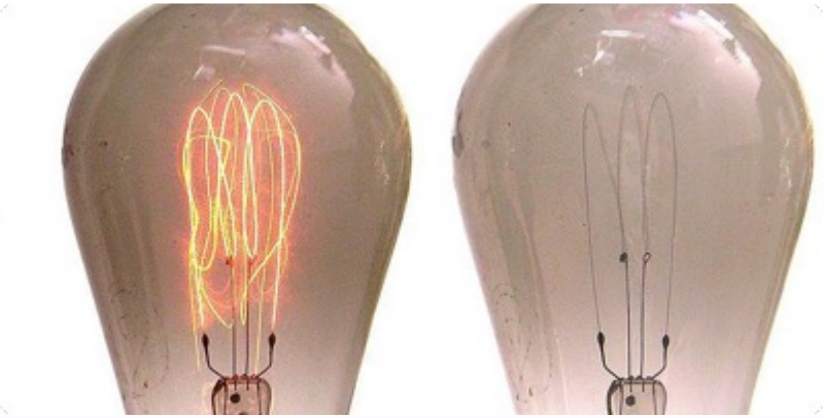
Halogen Lamps

- operate at higher temperature than a standard gas filled lamp of similar power without loss of operating life

- higher efficacy (10–30 lm/W)

Why Halogen …
Ordinary incandescent lamps
– tungsten is mostly deposited on the bulb

Halogen gas + Tungsten filament
>> chemical reaction – a *halogen cycle*
- increases the lifetime of the bulb
- prevents its darkening by redepositing tungsten from the inside of the bulb back onto the filament



Halogen Cycle

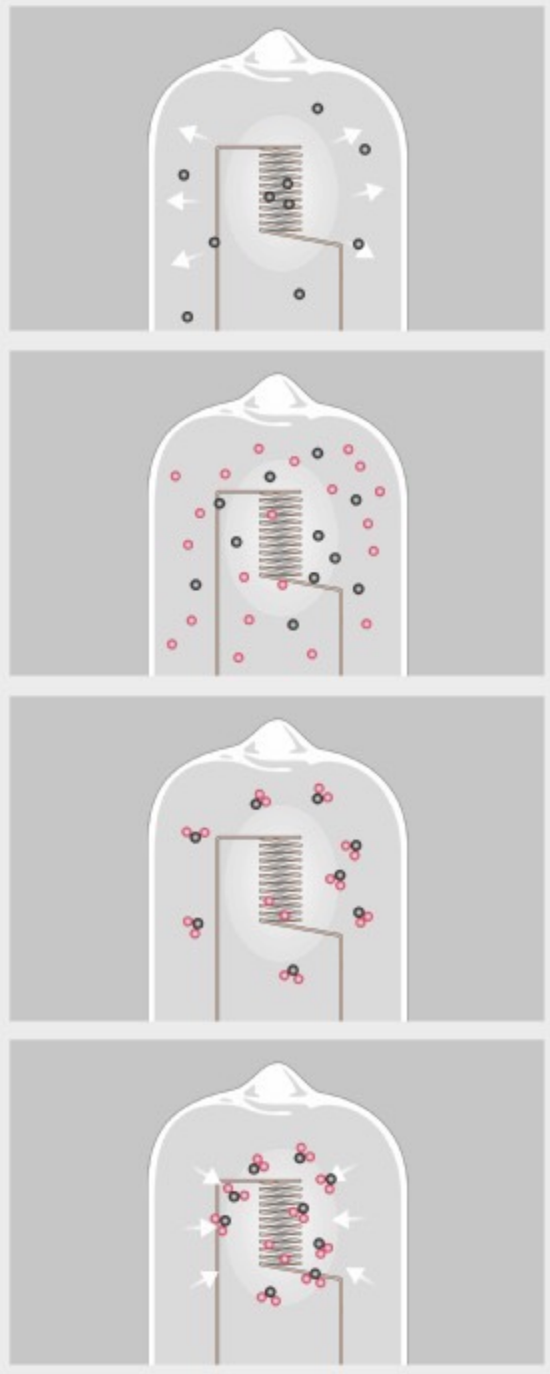
- Tungsten atoms evaporate from the hot filament
- diffuse near cooler bulb wall
- filament temperature ~ 3030º Celsius; bulb wall ~730º C

- Tungsten, oxygen and halogen atoms combine near the glass
- form tungsten oxyhalide molecules
- Halogen used: originally iodine, now bromine

- Tungsten oxyhalides remain in a vapor phase at the bulb-wall
- vapor moves toward the hot filament
- diffusion and convection currents are responsible for the movement

- High temperatures near filament break the tungsten oxyhalide molecules apart oxygen and halogen atoms move back toward the bulb wall
- tungsten atoms re-deposit on the filament

- cycle repeats

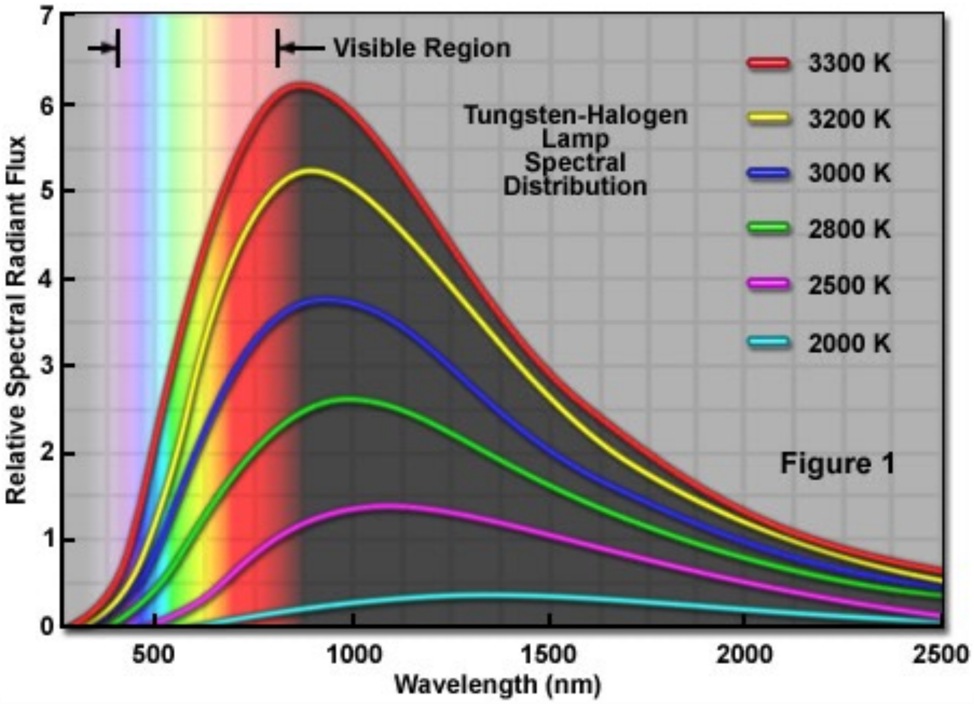


Halogen Lamps

- produces a continuous* spectrum of light
- from near ultraviolet to infrared

  (*c.f.* arc lamps – “spikey” spectra)
- filament operates at higher temperature than a non-halogen lamp
  - the spectrum is shifted toward blue

  = higher effective colour temperature



Halogen Lamps

Handling

- contamination – notably fingerprints – can damage the quartz envelope when it is heated
- contaminants create a hot spot on the bulb surface
- extreme, localized heat causes the quartz to change from its vitreous form into a weaker, crystalline form that leaks gas
- may form a bubble
- leading to failure or explosion

∴ quartz lamps should be handled without touching the clear quartz

Power, Supply

Power requirements –
basic Ohm's Law and Power equations

P = **VI; Power (Watts)** = Volts x Amps
V= IR; Volts = Amps x Ohms

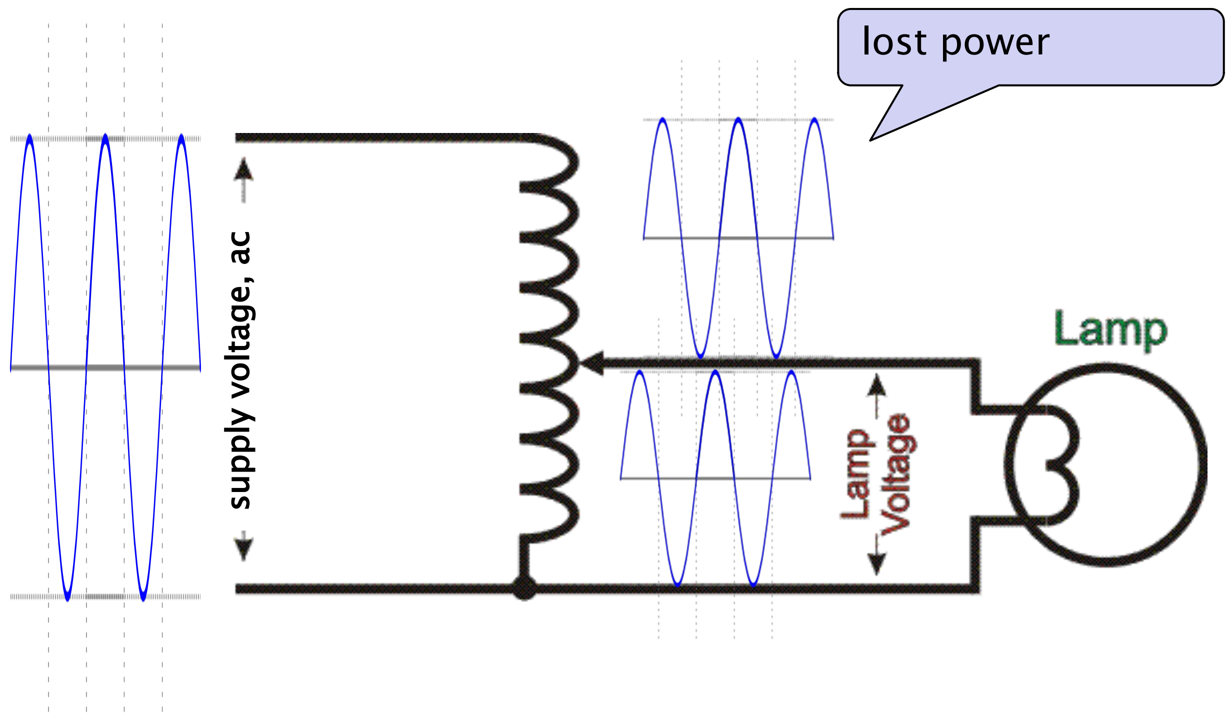
note – for generators,
NET POWER** quoted as **kVA
POWER in the load is kW
... due to ***Power Factor***

Current protection, Fuses: use the next available current rating up;
although circuits in professional studios and sets will be protected by **RCDs**

RCDs – monitor for leakage current (in microamps) in the earth circuit

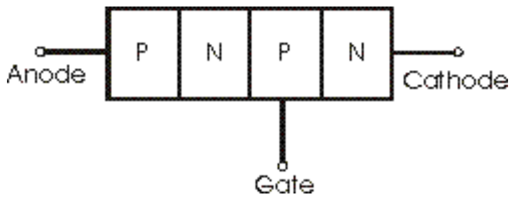
Standard voltage supply (UK) is 230V (+15% – 10%) ...
although studios will often have three phase power supply ...

Variac

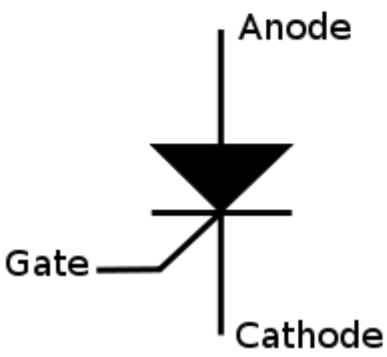


(From 1934 to 2002, *Variac* was a U.S. trademark of General Radio)

Solid State Control Devices



Typically: Thyristors – **Silicon Controlled Rectifiers – SCRs**

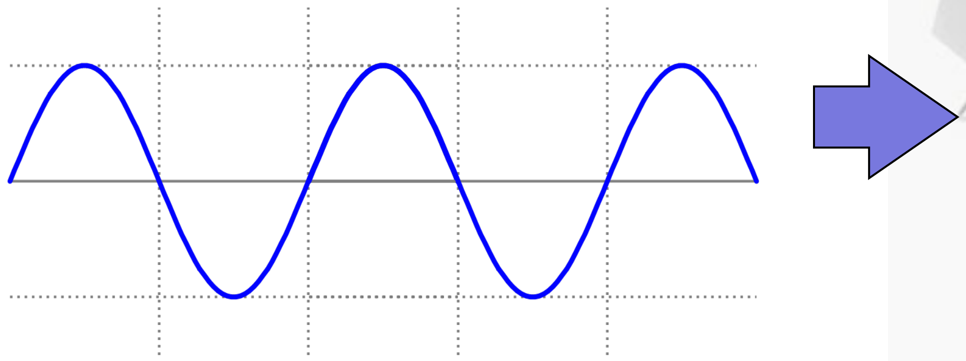


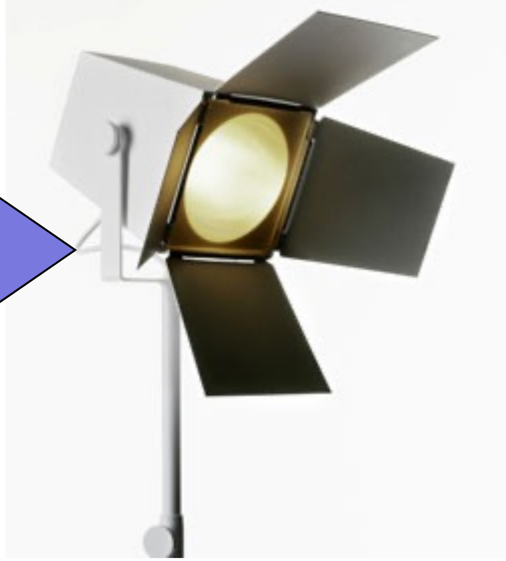
An SCR applied a "controlled rectification" of the wave form

- the device can be set to turn on at specific points of the wave form
- thus a modified supply wave is allowed through to the device
- i.e. the load receives less than full power
- Gate signal controls the current flowing through the device

Consider AC supply for a lamp:, full power

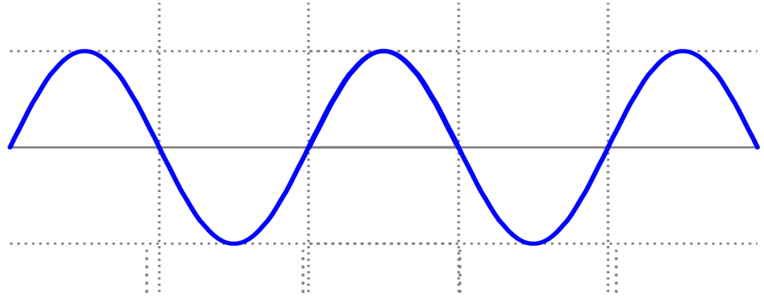
- all of the power is contained in the AC wave form as it passes through each 360º cycle



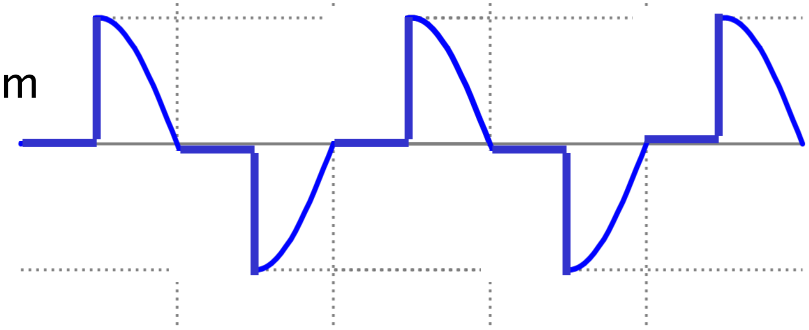


Solid State Control Devices

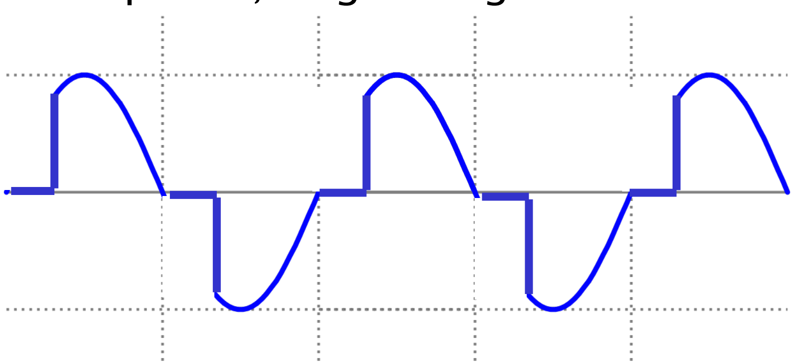
Full waveform



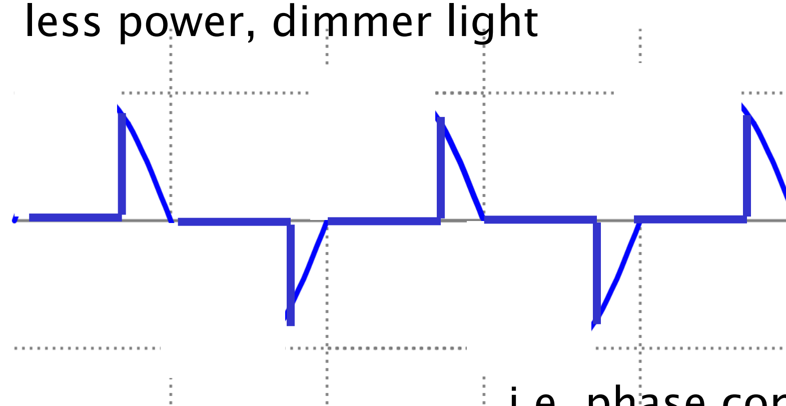
Controlled waveform



Cutting the wave off earlier = more power, brighter light



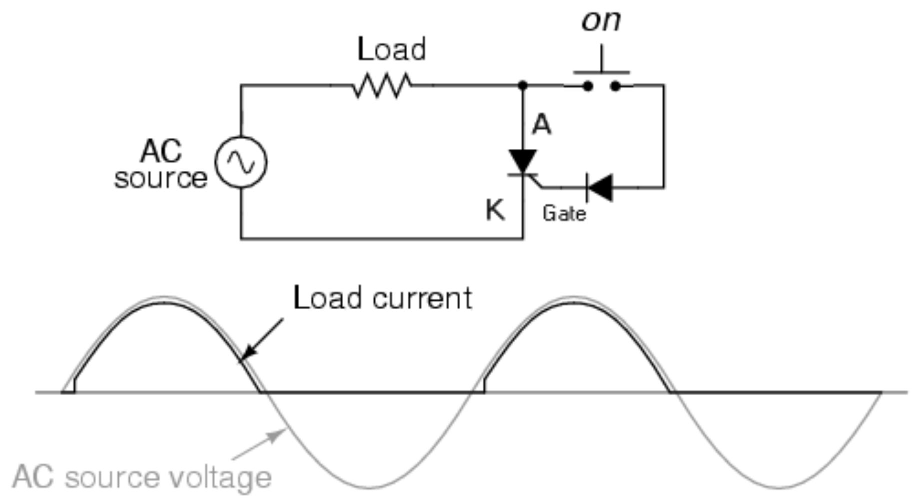
Cutting the wave off later = less power, dimmer light



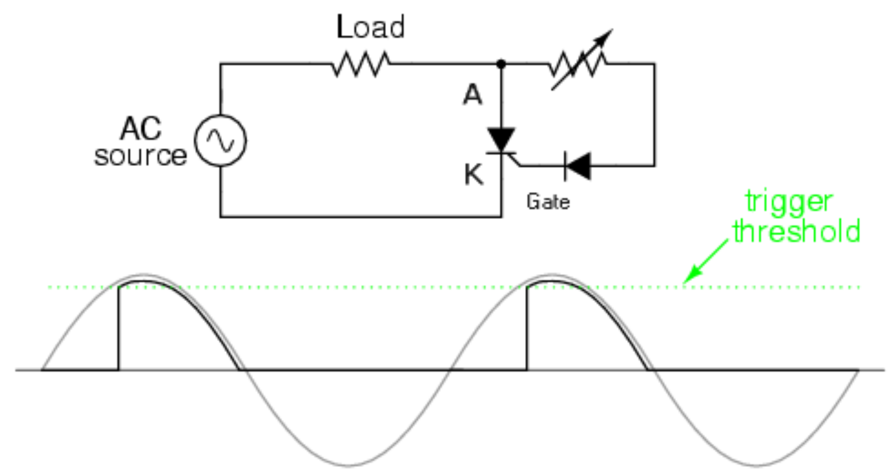
i.e. phase control

Solid State Control Devices

SCR applied – for one half of the waveform

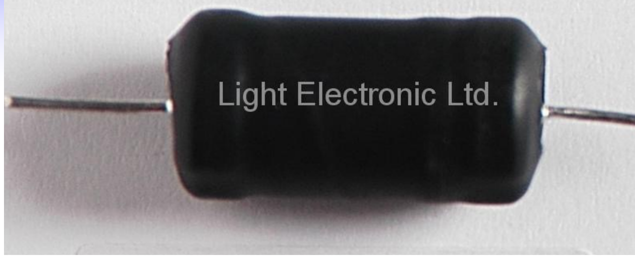


Circuit for applying power to an AC load (e.g. a lamp)



By building the potentiometer into the circuit, control gate into the SCR allows the cut-off point to be set

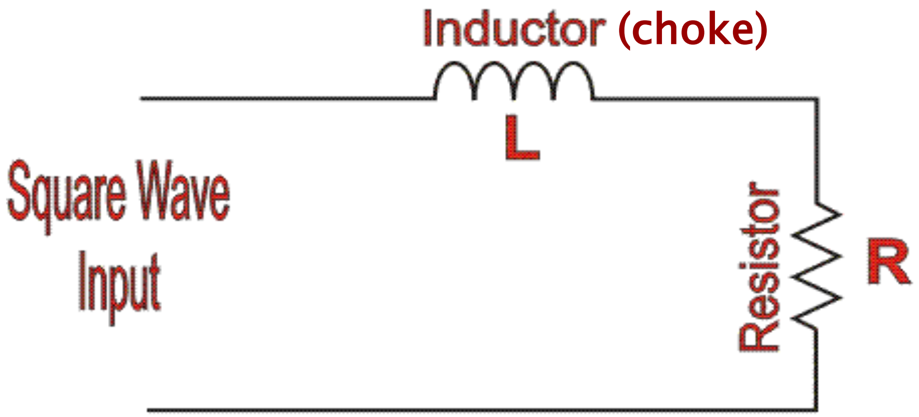
Buzzing Problems with Dimmers

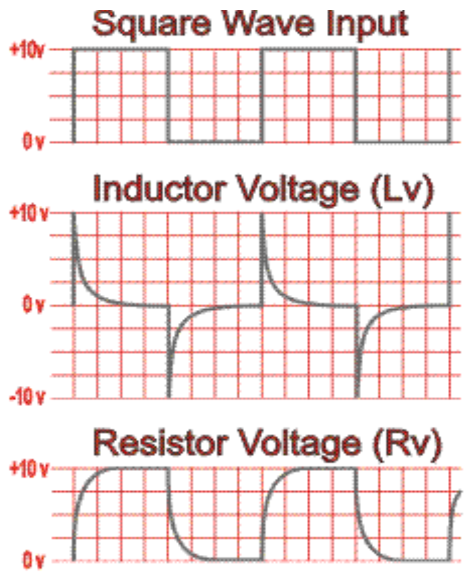


Choke

Dimmer – normally has a filter choke

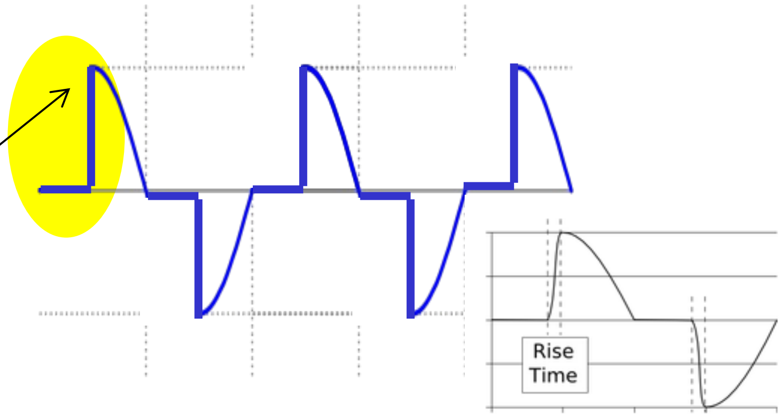
- chokes filter out electrical noise



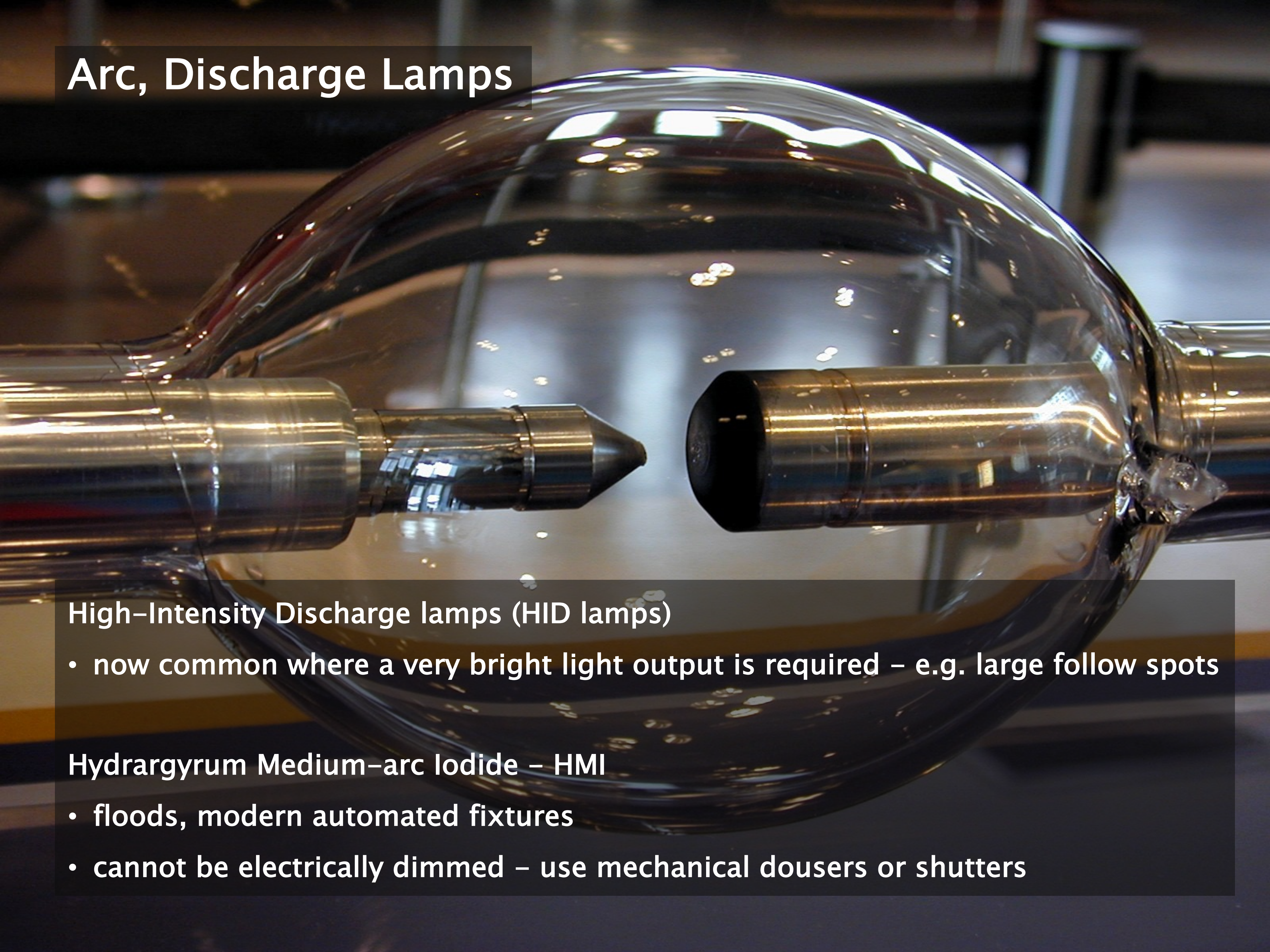


Square wave has "hard edges" consisting of harmonics
The choke (inductor) takes all this immediate rise, leaving a softer rise of voltage to the resistor

For the waveform supplied by a **SCR**, the rapid switch time also requires a choke filter



Arc, Discharge Lamps



High-Intensity Discharge lamps (HID lamps)

- now common where a very bright light output is required – e.g. large follow spots

Hydrargyrum Medium-arc Iodide – HMI

- floods, modern automated fixtures
- cannot be electrically dimmed – use mechanical dousers or shutters

Arc, Discharge Lamps

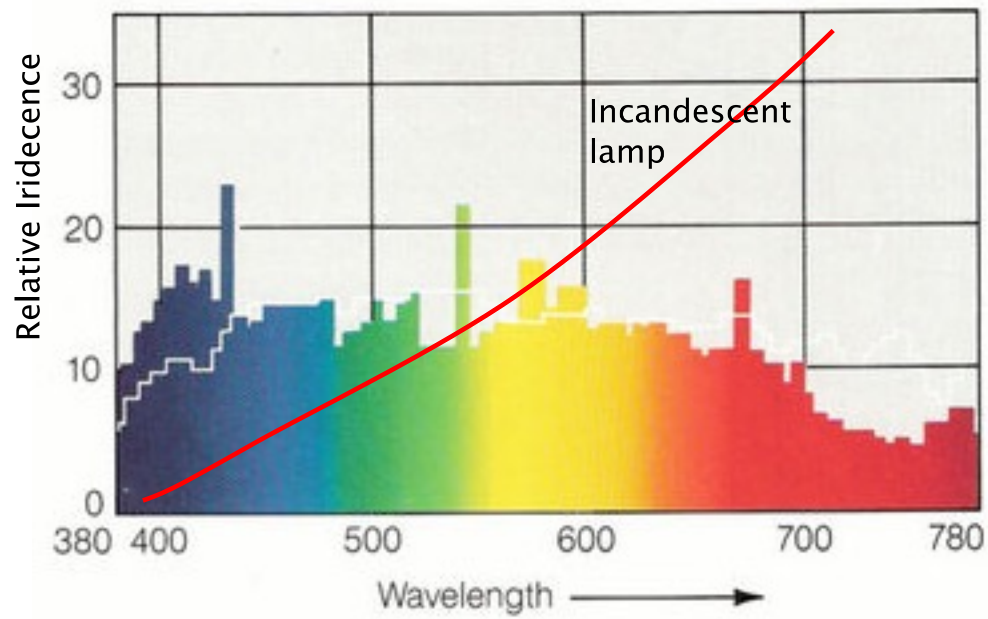
- electric arc in lamp of gas
- gas ionized by a high voltage
- therefore becomes electrically conductive

To start

- small voltage lamps – no problem – relatively small voltage to strike the arc
- Larger lamps:
  - very high voltage is pulsed across the lamp to "ignite" or "strike" the arc across the gas
  - requires an electrical circuit with an igniter and a ballast
  - ballast – is wired in series with the lamp

Colour Temperature
HMI – Hydrargyrum Medium-Arc Iodide

- Incandescent lamp – peak irradiation lies much farther in the non-visible infrared area of the spectrum
- compare to the wavelength of the maximal irradiation of the HMI
- relative efficiency of HMI in producing visible light, whereas incandescent bulbs produce a much greater portion of non-usable heat.



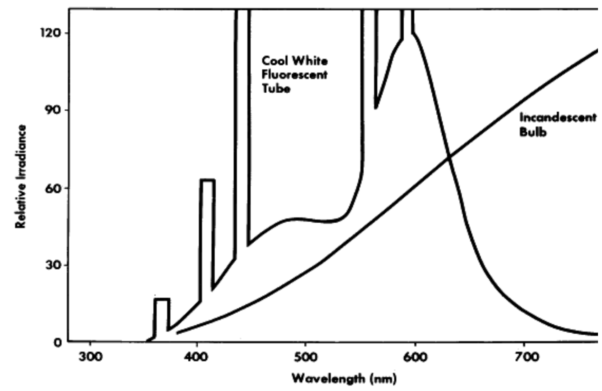
Spectral power distribution of a 575 W/GS OSRAM METALLOGEN HMI lamp and the spectral distribution of daylight at 5500K (White curve)

http://galaxylightingrepair.com/images/hmi_info.jpg

Colour Temperature - Correlated Colour Temperature

Colour Temperature for 'spiky' source

- need to match arc source to incandescent source
- then take the colour temperature of the incandescent source

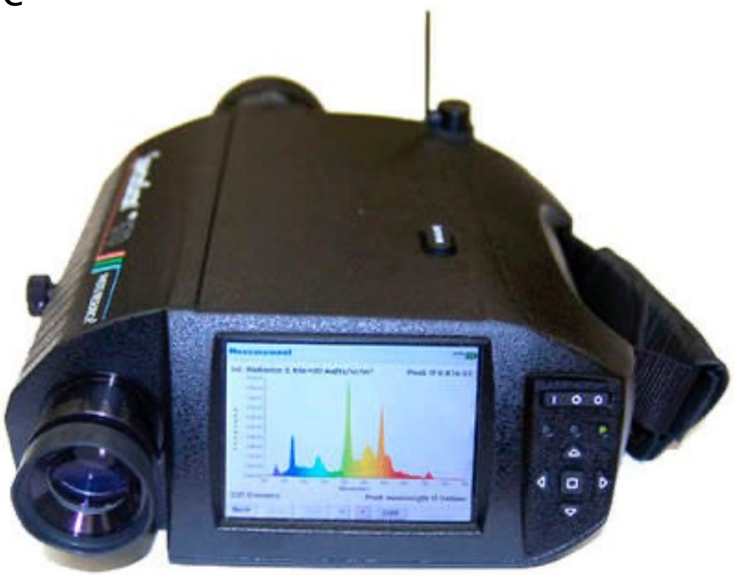


However - to match colours for the convenience of the human eye ...

... not convenient - so meters used

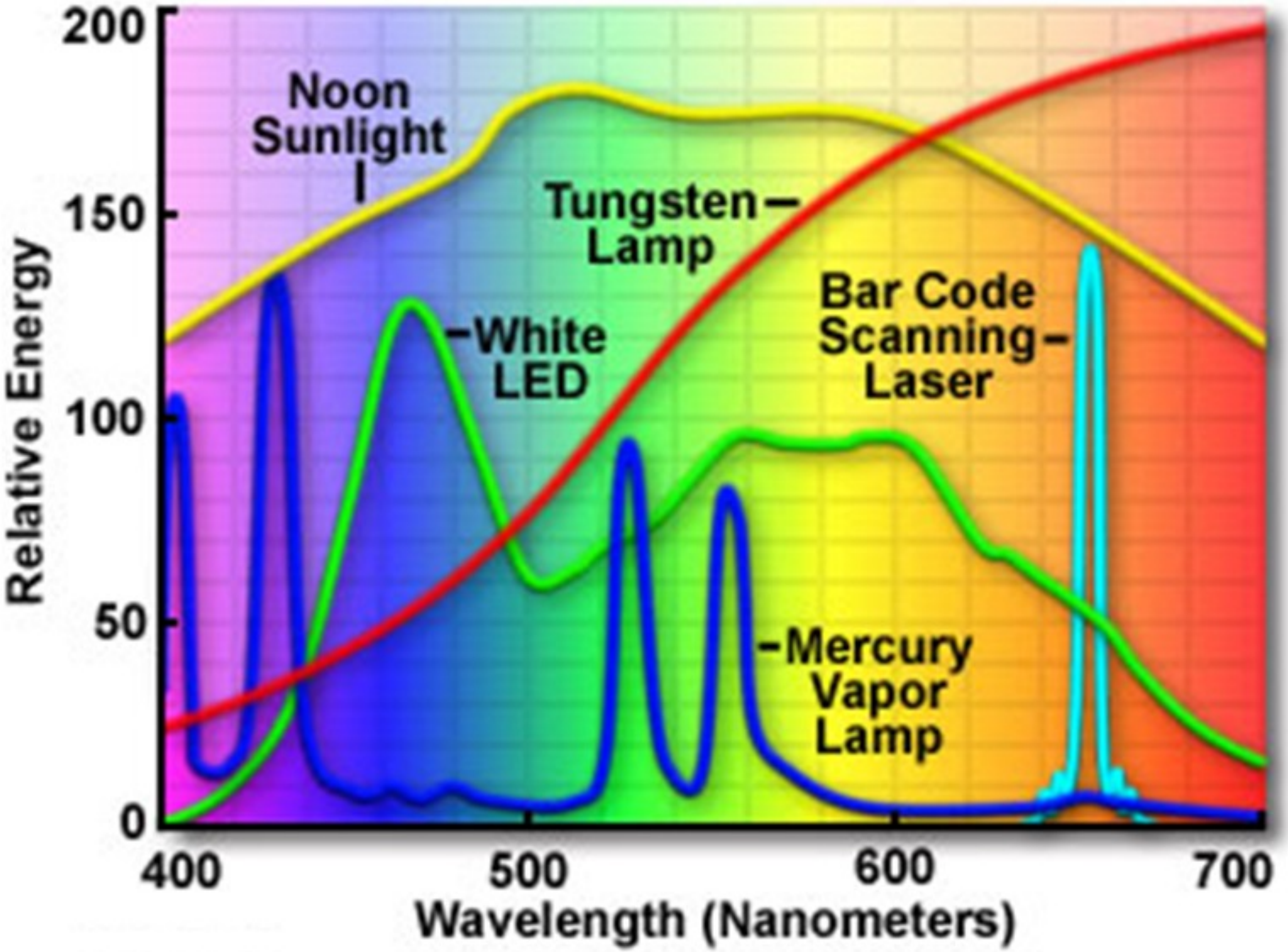
- meters designed to match the response of the eye
- but difficult to predict response of the human eye or TV camera
    - spectroradiometers
        - made by - Thoma, Photo Research and Gamma Scientific

          ... analyze peaks and troughs



PR-670 SpectraScan

Spectra of Common Sources of Visible Light



Mired

Micro Reciprocal Degree

- to convey colour temperature
- direct reciprocals of a corresponding degrees Kelvin colour temperature
- simplifies working with colour temperatures
- determine the strength of correction filter or gel

$$M = \frac{1{,}000{,}000}{Colour\ Temperature,\ Kelvin}$$

http://lacolouronline.com/blog/?10171-Defending-4500K-colour-Temperature

Mired Shift

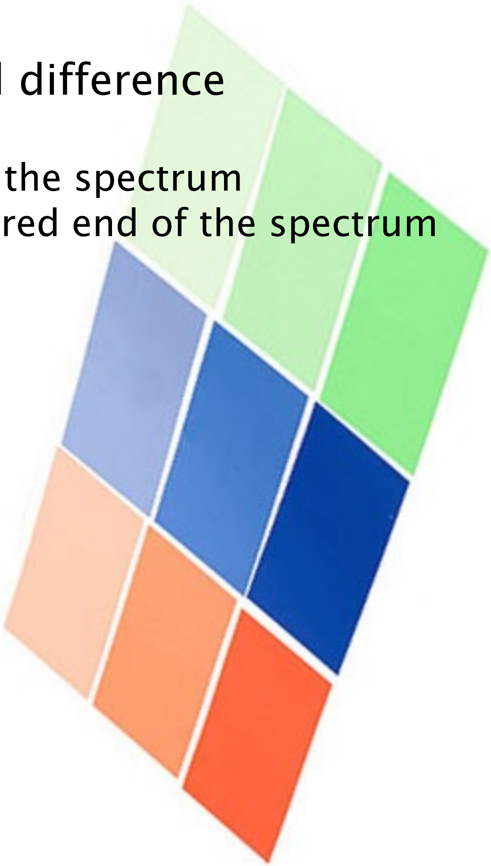
Calculate required correction gel

- 'shift' a light source from (i) original colour temperature to (ii) a new one

Desired colour temperature – Original colour temperature
= mired difference

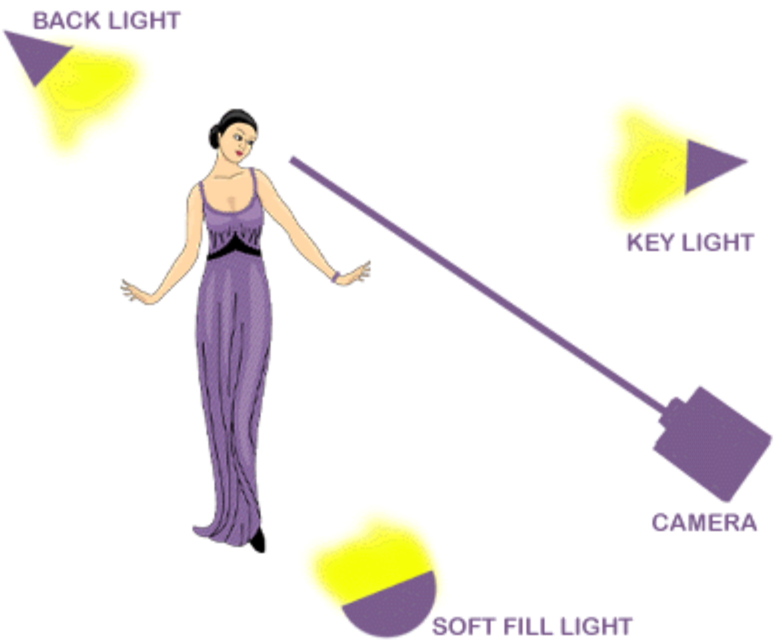
... so requires a colour correction filter equal to the mired difference

- negative: increases colour temperature towards the blue end of the spectrum
- positive: decreases the colour temperature towards the orange/red end of the spectrum



http://lacolouronline.com/blog/?10171-Defending-4500K-colour-Temperature

Mired Example



- Key light is metered as 2900K

$$= \frac{1{,}000{,}000}{2900} = 344.8\,\mathit{Mireds}$$

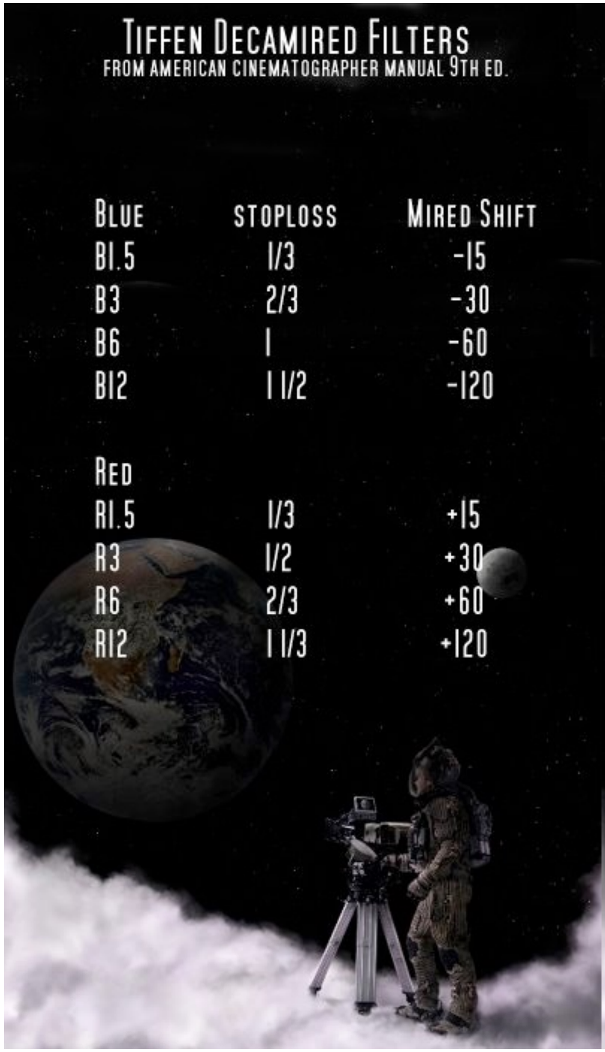
- but colour temperature required is 3200K

$$= \frac{1{,}000{,}000}{3200} = 312.5\,\mathit{Mireds}$$

312.5 - 344.8 = **−32.3** ***Mireds***

- *round down to* ***−32***
- ***negative*** *means* ***increase*** *colour temperature* ***to blue*** *end of spectrum*
- *CT Blue filter*

Mired Filters



Tiffen Decamired Filters
From American Cinematographer Manual 9th Ed.

Blue	Stoploss	Mired Shift
B1.5	1/3	-15
B3	2/3	-30
B6	1	-60
B12	1 1/2	-120
Red		
R1.5	1/3	+15
R3	1/2	+30
R6	2/3	+60
R12	1 1/3	+120

Lighting Filters for Color Temperature
From American Cinematographer Manual 9th Ed.

Increase Color Temperature (Blue)	Mired Shift
Lee #200 Double CT Blue	-274
Rosco #3220 Double CT Blue	-274
GAM #1520 Extra CT Blue	-190
Formatt #200 Double CT Blue	-274
Lee #201 Full C.T Blue	-137
Rosco #3202 Full Blue	-131
GAM #1523 Full CT Blue	-141
Formatt #201 Full CT Blue	-137
Lee #281 Three Quarter CT Blue	-112
Rosco #3203 Three Quarter Blue	-100
GAM #1526 Three Quarter CT Blue	-108
Formatt #281 Three Quarter CT Blue	-112
Lee #202 Half CT Blue	-78
Rosco #3204 Half Blue	-68
GAM #1529 Half CT Blue	-75
Formatt #202 Half CT Blue	-78
Rosco #3206 Third Blue	-49
Lee #203 Quarter CT Blue	-35
Rosco #3208 Quarter Blue	-30
GAM #1532 Quarter CT Blue	-38
GAM #1534 Sixth CT Blue	-28
Lee # 218 Eighth CT Blue	-18
Rosco #3216 Eighth Blue	-12
GAM #1535 Eighth Blue	-20
Formatt #218 Eighth CT Blue	-18

Lighting Filters for Color Temperature
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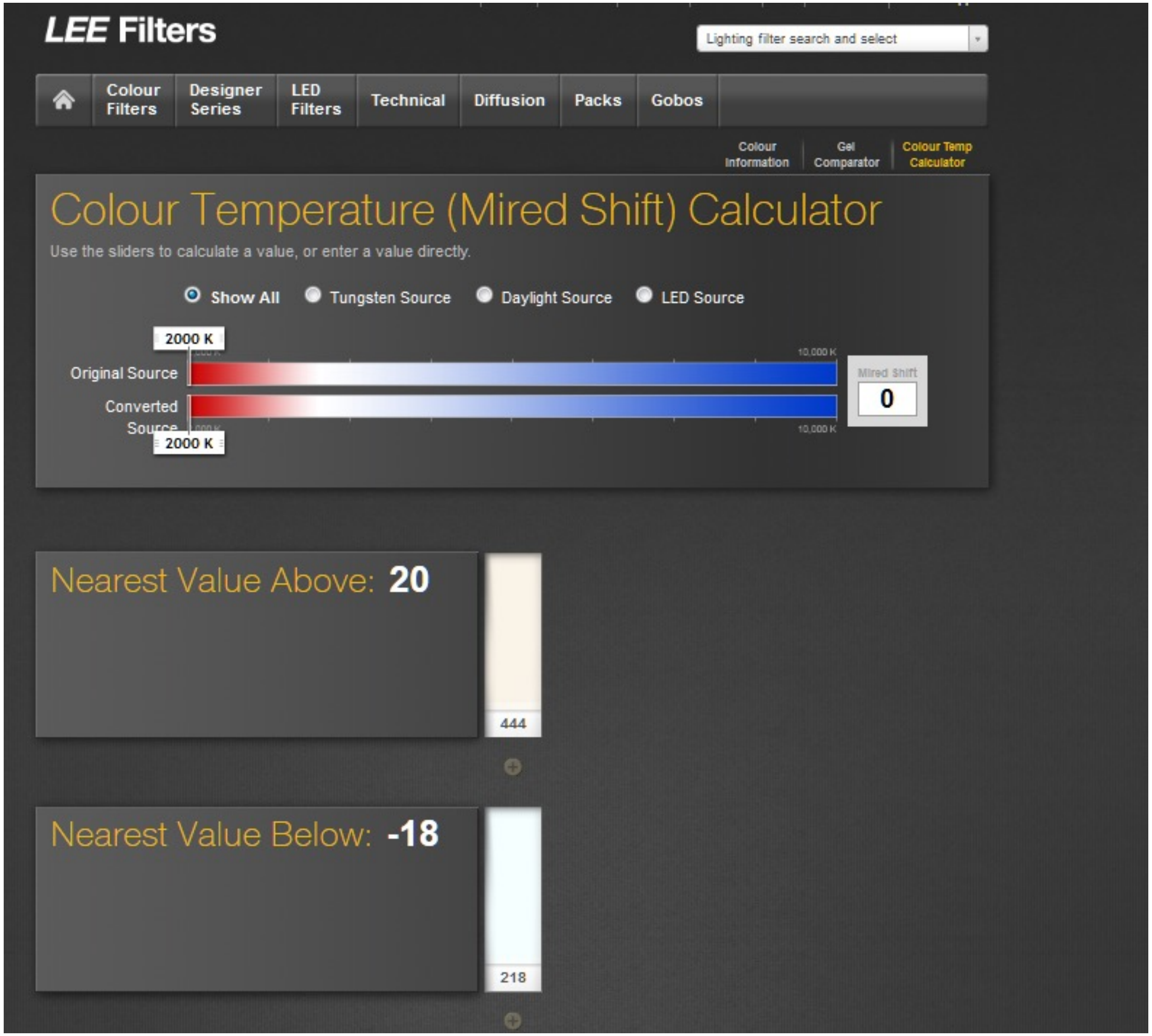
Formatt #281 Three Quarter CT Blue	-112
Lee #202 Half CT Blue	-78
Rosco #3204 Half Blue	-68
GAM #1529 Half CT Blue	-75
Formatt #202 Half CT Blue	-78
Rosco #3206 Third Blue	-49
Lee #203 Quarter CT Blue	-35
Rosco #3208 Quarter Blue	-30
GAM #1532 Quarter CT Blue	-38
GAM #1534 Sixth CT Blue	-28
Lee # 218 Eighth CT Blue	-18
Rosco #3216 Eighth Blue	-12
GAM #1535 Eighth Blue	-20
Formatt #218 Eighth CT Blue	-18

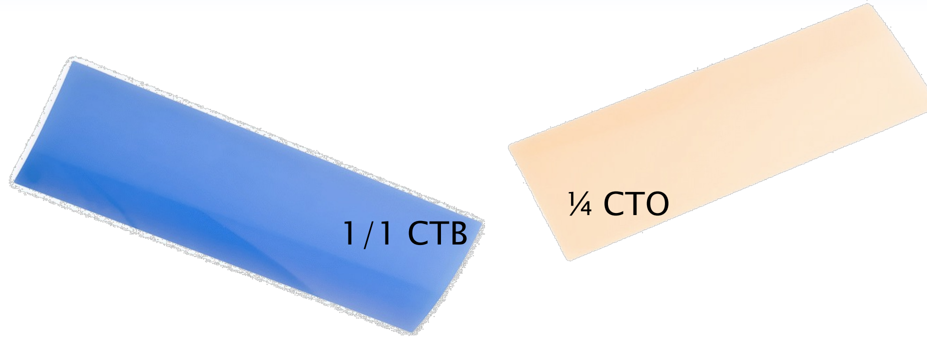
http://www.ryanpatrickohara.com/download/colourtemperature.pdf

Mired Filters

http://www.leefilters.com/lighting/mired-shift-calculator.html



Colour Balance



Colour correction gels

- CTB (colour temperature blue)
- CTO (colour temperature orange)

- CTB gel converts tungsten light of 3200K to 'daylight' colour
- CTO gel performs the reverse

- Beware different manufacturers' gels yield slightly different colours
- definition of the colour of daylight – varies depending on the location (latitude, dust, pollution) and the time of day

Fluorescent lights

- have a green cast – removed by a **minus green gel**
- to add green cast – **plus green**

Strength of Gels

- 3/4, 1/2, 1/4, and 1/8
- A 1/2 CTO gel is half the strength of a (**full**) CTO gel

LCD with minus green

Colour Balance

Colour	Name	Effect
	Cyan	Absorbs Red
	Yellow	Absorbs Blue
	Magenta	Absorbs Green
	Red	Absorbs Blue and Green
	Green	Absorbs Red and Blue
	Blue	Absorbs Red and Green

(colour correction is more subtle than shown here)

Colour Balance

Filters stop light from passing through them, so the resultant light is reduced

- colour correction gels only reduce small amounts of colour so are fairly "pale" filters
- light reduction is minimal (but definitely there)

Tungsten Light Source Correction

- CTB – convert Tungsten Light sources of 3200K to colour, comparable to daylight
- make a key light appear to be from outdoor source such as the sky or a window

201 Full CTB : Tungsten to Daylight 3200K – 5700K
202 Half CTB : Tungsten to Daylight 3200K – 4300 K
203 Quarter CTB : Tungsten to Daylight 3200K – 3600K

219 Fluorescent Green: convert tungsten light to general fluorescent lighting
242 Fluorescent 4300K: convert tungsten light to fluorescent @ 4300K

http://www.onstagelighting.co.uk/lighting-design/colour-correction-filters-stage-lighting-gels/

Colour Balance

Daylight Light Source Correction

- CTO – convert blue-ish light source of daylight to a range of tungsten colour temperatures
- useful for balancing light sources from a window with tungsten key lighting
- also used with a tungsten light to create a warm sunny keylight.

204 Full CTO: Daylight 6500 K to Tungsten 3200 K
205 Half CTO: Daylight 6500 K to Tungsten 3800 K
206 Quarter CTO: Daylight 6500 K to Tungsten 4600 K

Other filters for different light sources

- HMI , CID and CSI arc lamps
- some common correction filters for specific uses:

236 HMI to Tungsten 3200K: warm orange tint that can be used for sunlight in tungsten lights
237 CID to Tungsten 3200K: Light red/brown. Used with tungsten, produces a outdoor summer mood
238 CSI to Tungsten 3200K: similar to 237CID

LED Lighting

BBC Breakfast TV Studio, Media City, Salford
includes LED lighting

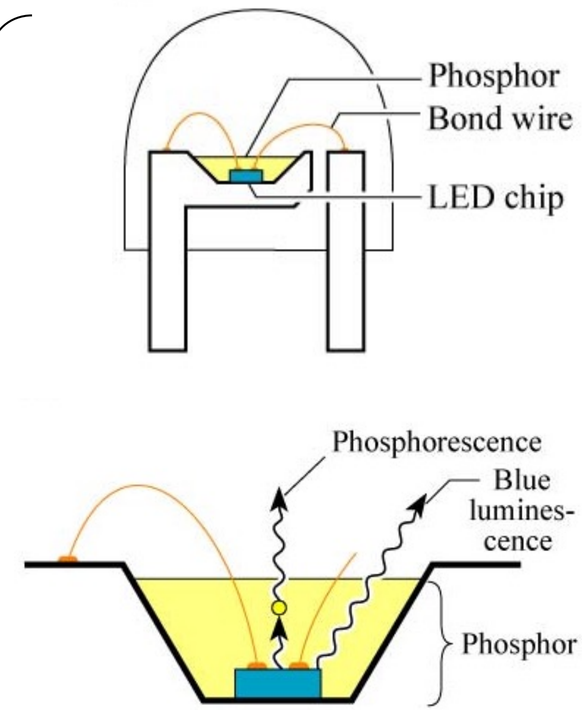
LEDs … and producing white light

(1) RGB LEDs

- combine red, green and blue LEDs
- mix for wide range of colours, including white
- RGB white light – generally poor
- possible to shape the spectral emission
  - but dimensions bigger, less beam control

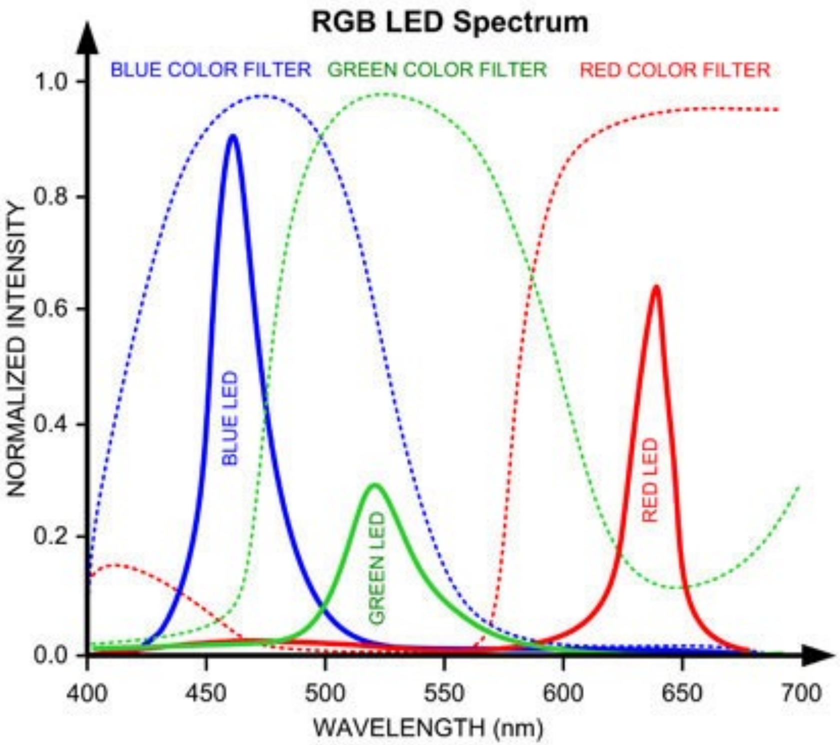
(… or 2 …) Luminescence Conversion

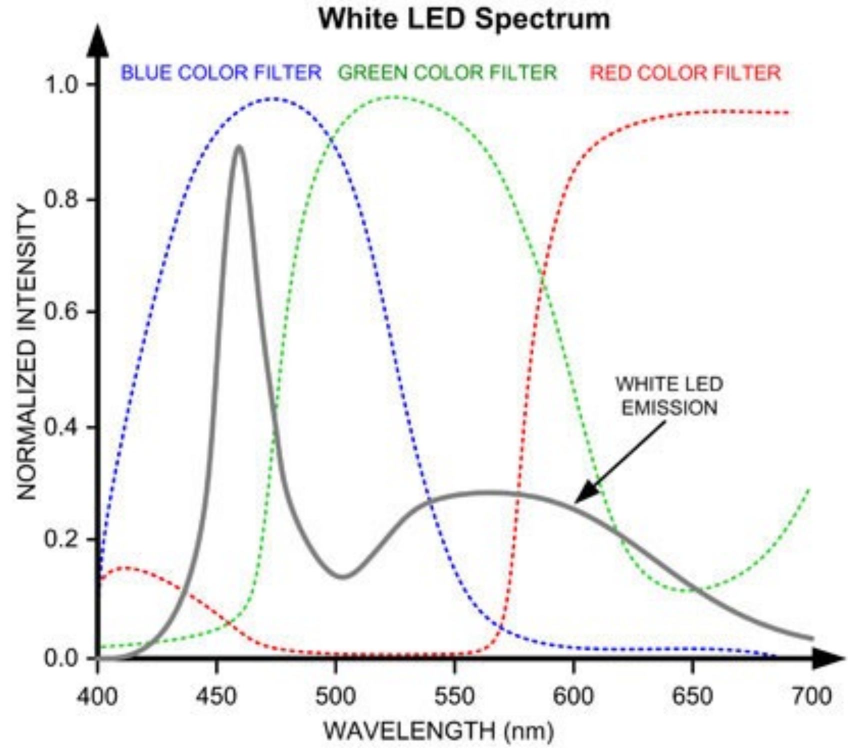
- convert the spectrum of a LED by using phosphors as a luminous layer
  - usually – blue LEDs with yellow phosphors coatings
- better spectral distributions, good colour rendering
  - warm white, neutral white, daylight white
    - colour temp 2500K to 8000K



(RGB and White LED sprectra – see next slide)

LED Spectra





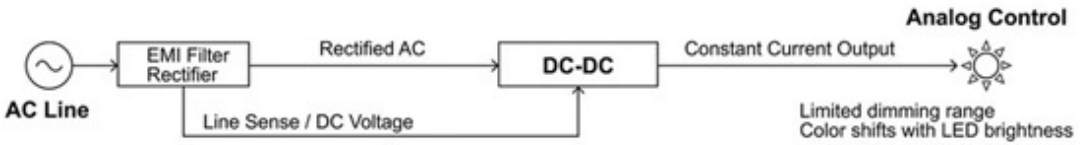
Typical colour filter spectra :
(left) RGB-LED spectrum has peaks for red, green and blue

(right) White-LED spectrum has dominant peak at 450nm and a lower peak at 600nm

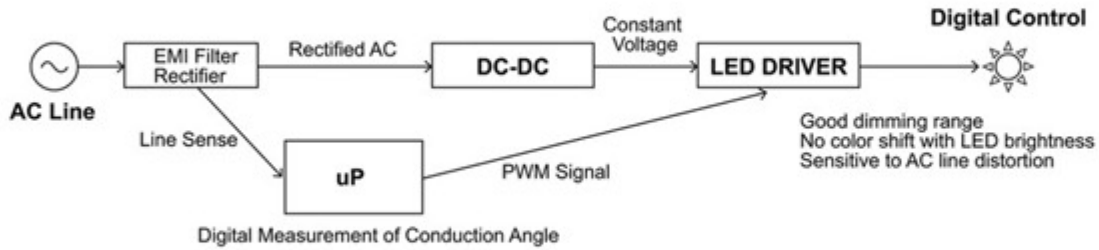
http://ideatoreality.wordpress.com/2010/04/25/artificial-lighting/

LED Control

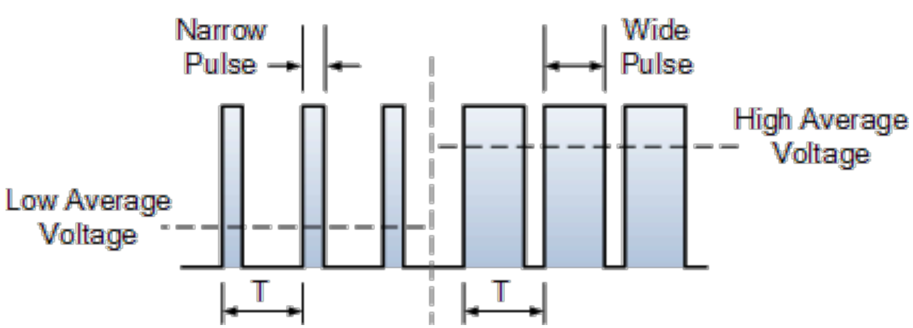
Analogue Controlled LED Dimming



Digital Controlled LED Dimming



Pulse Width Modulation (PWM)



- **First dimmable LED used analogue control**
- Sensed the rectified DC value of line voltage
- Directly control the output current of power supply
- Still popular for consumer LED lighting
- Dimming range is limited
- Colour shifts as current to leds is reduced

- **Recent systems - higher power LED lighting**
- Demand for better dimming characteristics
- Analog controls for LED drivers developed
- Improved efficiency
- Eliminated color shift with changes in brightness
- Pulse width modulation - PWM – for control signal signal

https://www.led-professional.com/resources-1/articles/solving-the-phase-cut-dimming-challenge

Moving Head Lights (DMX controls)

'Intelligent Lights'

- remotely controlled through a separate signal cable
- two cables – power + control information

Controlled from lighting desk

- colour changing
- multiple gobos
- prism effects
- beam shaping

Cost

- plumetted in the past few years but still cost £2k–3k

Speed

- Moving a heavy light through 360° is technically difficult; requires good design and large motors
- speed of movement is not very fast
- for lightning quick pans and tilts, a scanner is the only real option

Set up

- fairly easy but needs rigging – heavy light, setting DMX address, patching on desk

Martin Mac 500 profile light

Mac 2000, Scanner

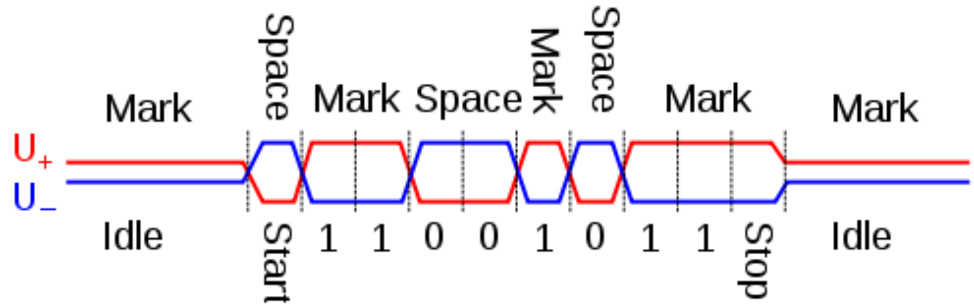
Martin Mac 600 fresnel light

DMX512 – Lighting Control Protocol

Standard for data networks to control stage lighting and effects

- differential signaling
- variable-size, packet based protocol
- one-way signals – no automatic error checking and correction

- **Not** for safety-critical applications e.g.
  - pyrotechnics
  - laser lighting displays
  - where audience or performer safety is at risk.

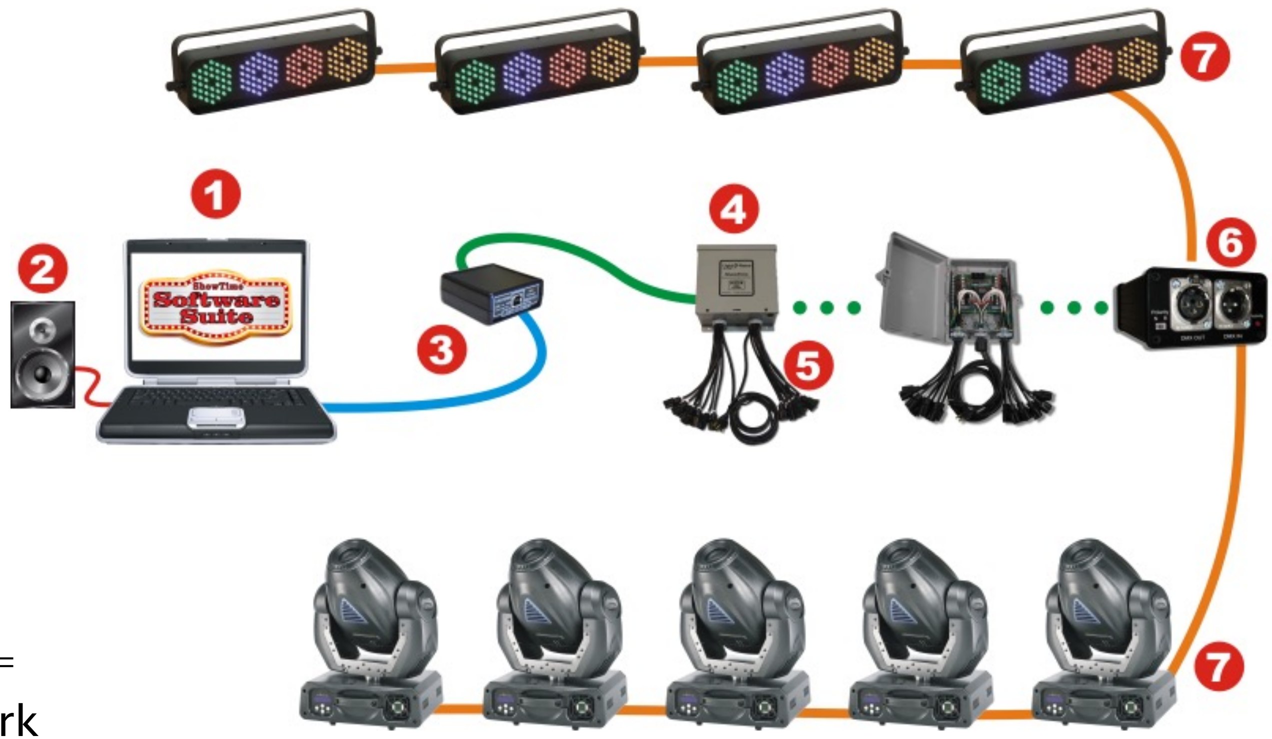


DMX512 – Lighting Control Protocol

Network topology

- multi-drop bus topology, daisy chain
- network implementation – single DMX512 controller + one or more controlled fixtures

- e.g. lighting console = controller for a network of (e.g.) dimmers, fog machines and intelligent moving lights



More details – see – http://www.lightorama.com/TypicalSetups.html#Simple_Controller_Configuration_Connected_to_DMX-512_Universe

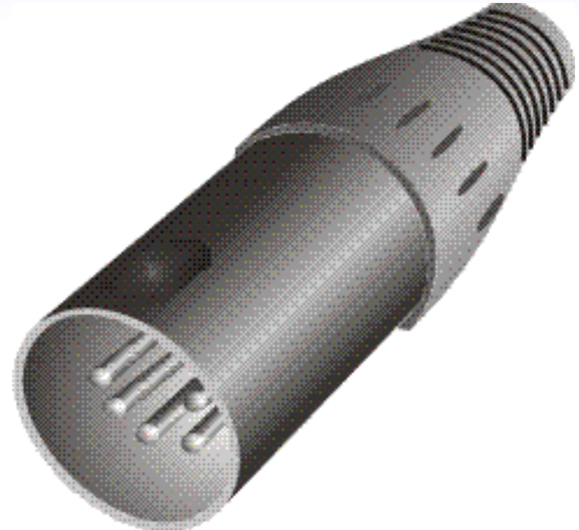
Each DMX network is called a "DMX universe"

- control desks may have the capacity to control multiple universes
- OUT connector for each universe

DMX512 – Lighting Control Protocol

- receiver fixture has
    - a DMX512 "IN" connector
    - and DMX512 "OUT" connector (sometimes marked "THRU")
- connections via DMX512 cable

DMX512 – Lighting Control Protocol



Connectors

- DMX512 1990 spec –
- data link – five-pin XLR-5 (5 pin)
- female connectors on OUT ports
- male connectors on IN/THRUports

XLR-5 pinout

1. Signal Common
2. Data 1– (Primary Data Link)
3. Data 1+ (Primary Data Link)
4. Data 2– (Optional Secondary Data Link)
5. Data 2+ (Optional Secondary Data Link)

RJ-45 (Ethernet connectors) are also used